

Toward Digital Twins for Fungal Computing

Viable XOR regimes, parameter identifiability, and waveform-guided calibration in a simulated excitable–memristive network

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CODE & DATA
fungal-computer-sim



FULL PAPER
PDF + appendices

96.0%

less waveform mismatch after refinement

Waveform matching improves rough ML guess into a better one

Mismatch **1.070** → **0.042**; core-parameter error **16.6%** → **8.8%** (−46.9%). One-sided Wilcoxon $p = 3.1 \times 10^{-5}$ and 6.1×10^{-5} , $n = 15$.

1 The substrate, the model, the probes

Mycelium is excitable and memristive, and has already been used to create Boolean logic. But **specimen-to-specimen variability** means an electrode layout that computes XOR on one specimen fails on the next — a digital twin would enable **calibration against a real specimen**.

RANDOM GEOMETRIC GRAPH · FITZHUGH–NAGUMO NODES · MEMRISTIVE EDGES

$$\tau_v \, dv_i/dt = v_i - v_i^3/3 - w_i + I_{stim,i} + I_{coup,i}$$

$$\tau_w \, dw_i/dt = v_i + a - b \, w_i$$

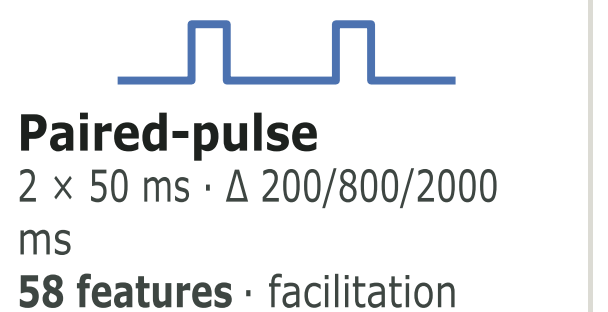
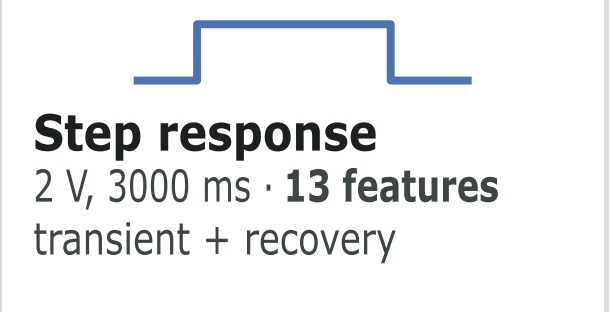
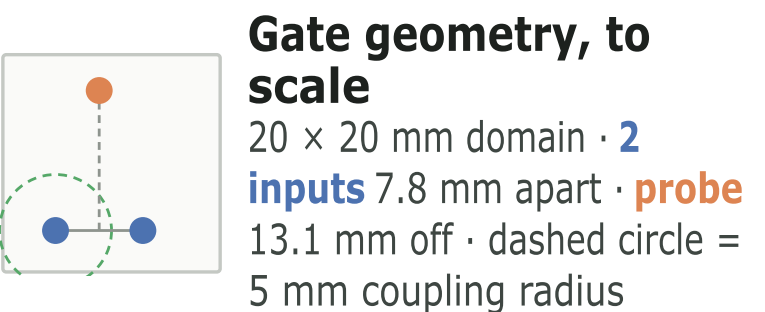
$$g_{ij} = 1 / [R_{on} m_{ij} + R_{off}(1 - m_{ij})]$$

$$dm_{ij}/dt = \alpha |v_i - v_j| (m_{ij}(1 - m_{ij}) + \Theta.05)$$

20 × 20 mm domain · 5 mm coupling radius · 20–120 nodes · **eight latent parameters**: $\tau_v, \tau_w, a, b, v_{scale}, R_{off}, R_{on}, \alpha$

$$S_{XOR} = \min(V_{10}, V_{01}) - \max(V_{00}, V_{11})$$

Separating margin. $S_{XOR} > 0 \Rightarrow$ one midpoint threshold gives 4/4 truth-table accuracy under noiseless readout; larger margins $\Rightarrow$ more tolerant gates. XOR is the test case because it is linearly inseparable.



160 optimization specimens → 400 characterization specimens → 94 features → random forest → 15 rediscovery specimens

3/8

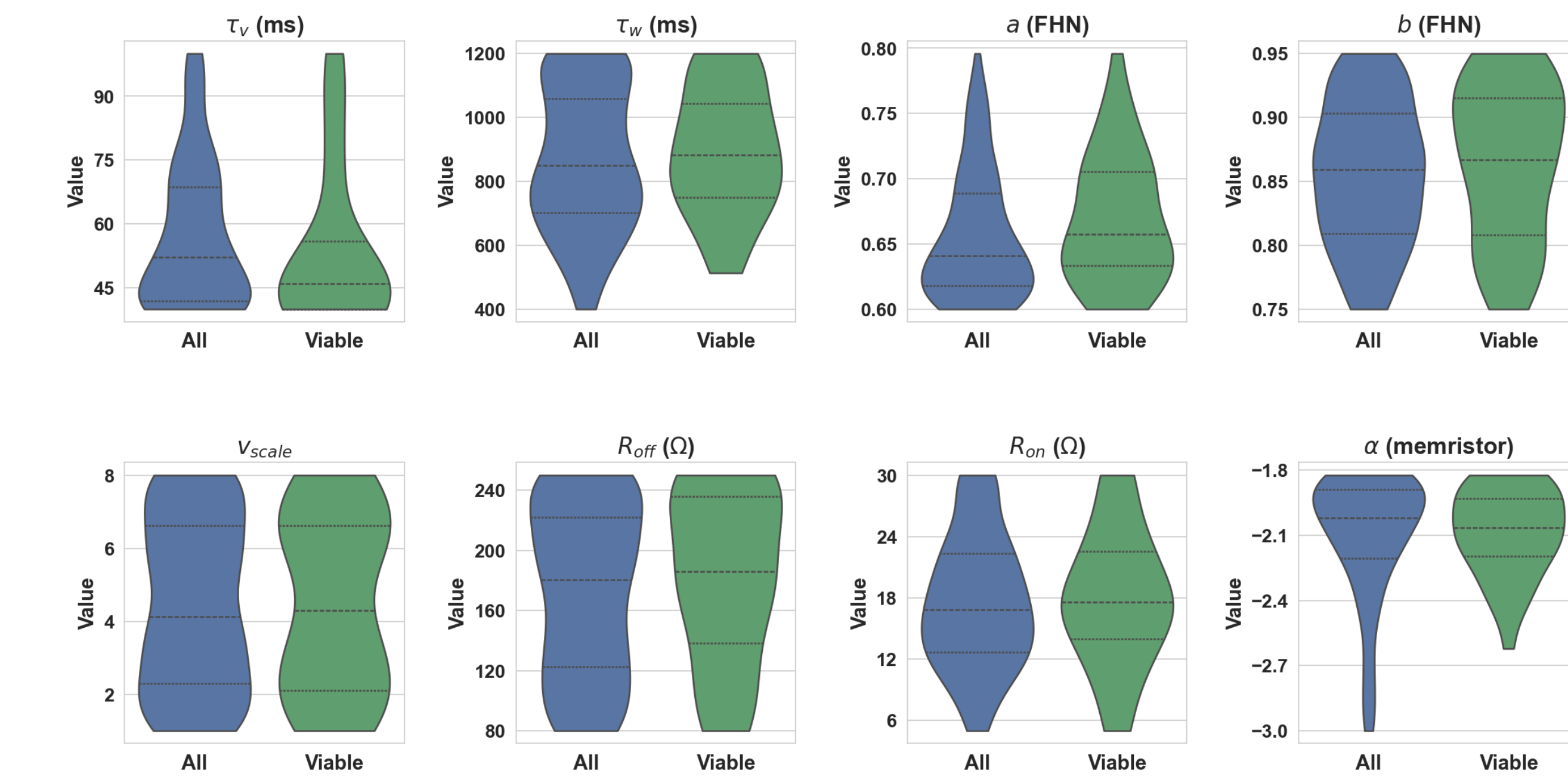
latent parameters reliably recoverable

Standard electrical probes give partial identifiability

Random forest $R^2 = \mathbf{0.912}$ (τ_v), **0.816** (τ_w), **0.717** (a). But $v_{scale}, R_{on}, R_{off}$ stay at $R^2 \leq 0.02$.

2 XOR viability is a constrained, multimodal subspace

Bayesian optimization tuned electrodes, voltage, pulse width and inter-input delay for 160 specimens, then tuned the eight parameters with geometry held fixed. **Viable = top quartile of tuned S_{XOR} (≥ 0.0627) → 40 specimens**, all 4/4. Below: **all 160** vs. **40 viable**.



56% / 60% of the τ_v / τ_w search ranges — the timescales are genuinely constrained.

34.6x threshold at best (2.172); viable median 0.148, IQR 0.090–0.399. Robust gates are rare.

4 of 8 marginals bimodal ($\tau_v, b, v_{scale}, R_{off}$): a family of regimes, not one optimum.

Geometry too: inputs 7.8 ± 6.4 mm apart, 13.1 ± 3.5 mm to output, 2.71 ± 0.91 V, 2869 ± 1634 ms.

Aligned

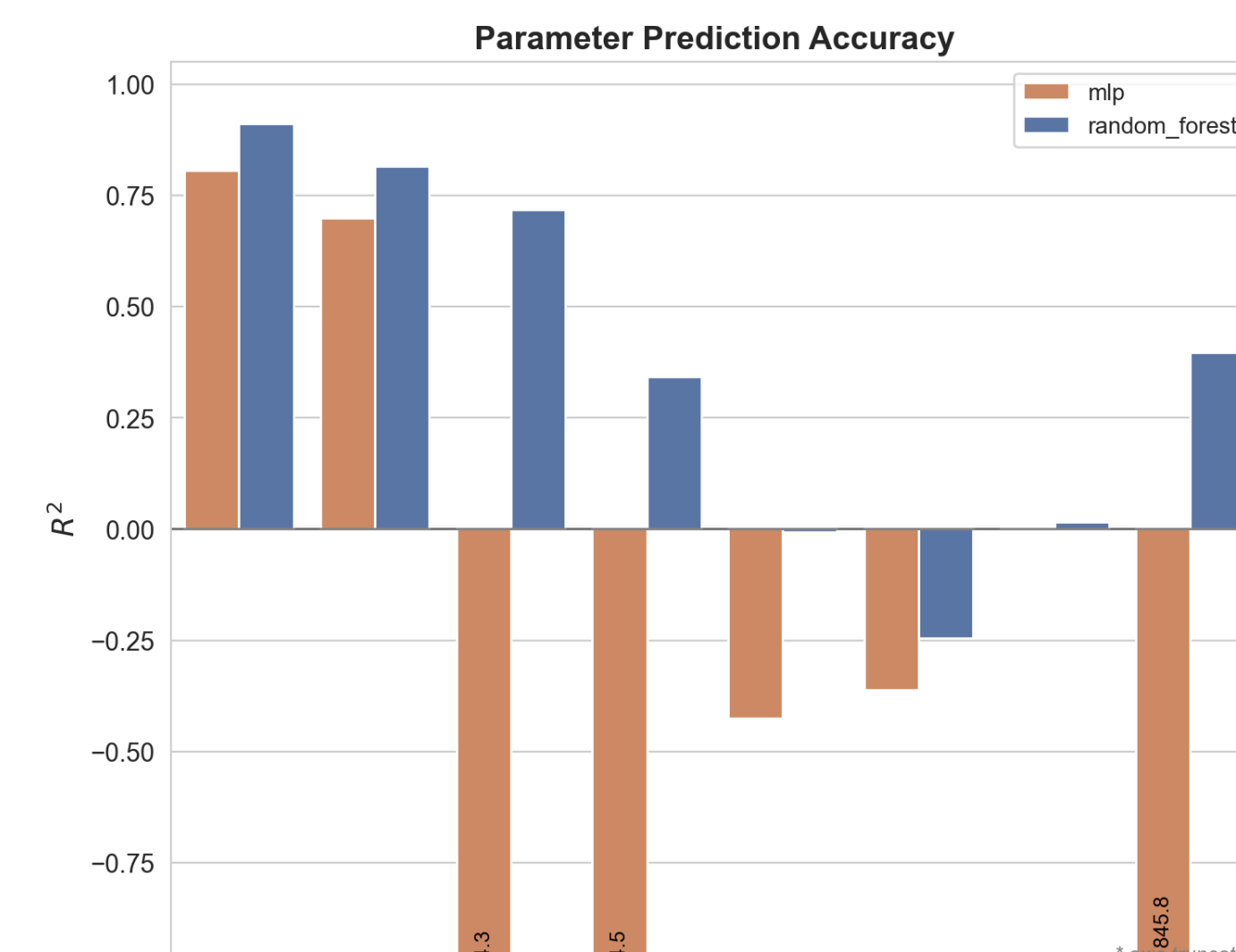
identifiability tracks functional importance

Identifiable parameters align with those that are most consequential for XOR behavior

The best-identified parameters are the most consequential; the worst-identified are the most forgiving. One exception — a — names the next protocol to build.

3 Probes recover some parameters, not all

400 specimens × 94 features, 80/20 split. **Random forests beat MLPs decisively** — the MLP is catastrophically negative on a, b and α (axis truncated) — so RF is used everywhere downstream.



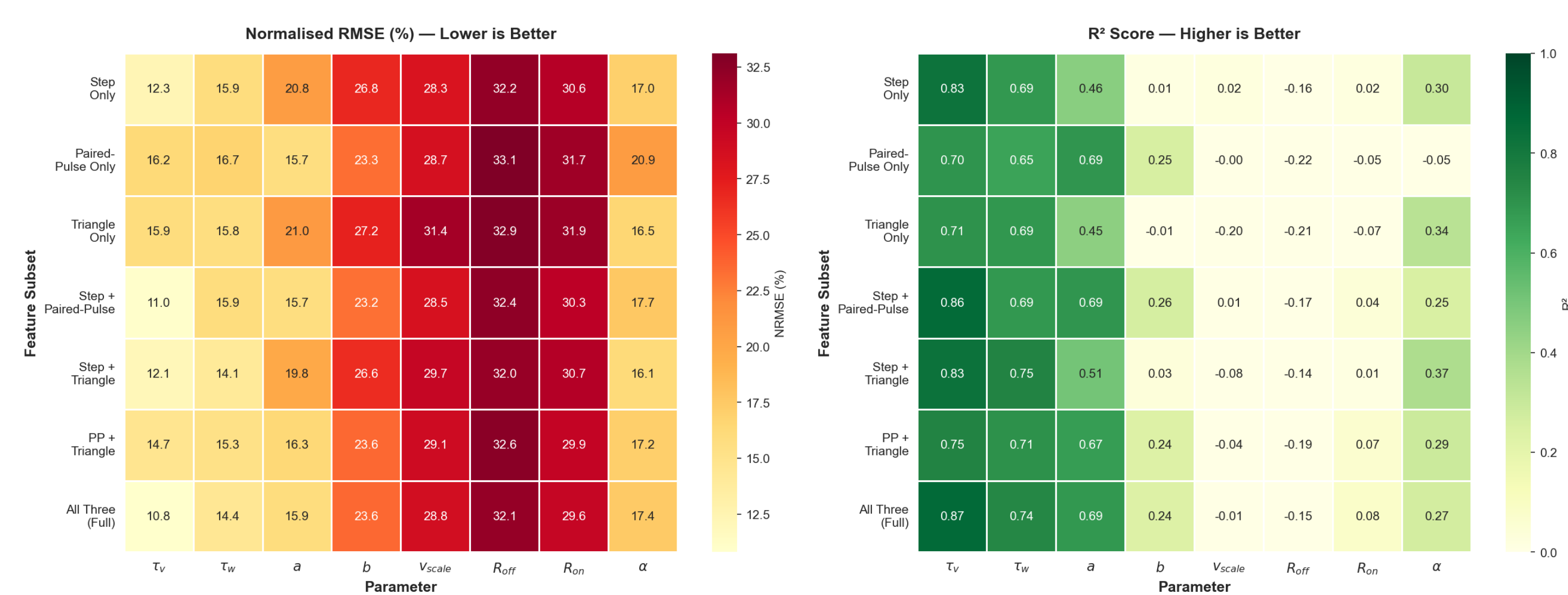
Held-out R^2 : **random forest** vs. **MLP**.

Param	R ² all	RMSE all	n	R ² viab.	RMSE viab.
τ_v	0.912	10.566	37	0.746	8.857
τ_w	0.816	145.757	58	0.730	127.935
a	0.717	0.044	47	0.065	0.047
b	0.342	0.068	63	0.210	0.056
α	0.396	0.003	30	−0.038	0.004
v_{scale}	−0.007	2.501	71	−0.031	2.218
R_{off}	−0.244	81.195	62	−0.196	63.808
R_{on}	0.017	13.538	39	−1.340	12.259

Held-out R^2 : performance overall and inside each parameter's viable interval.

Read RMSE, not just R^2 . Absolute error inside the viable interval is equal or lower than overall for six of eight parameters, even where viable-range R^2 collapses — range restriction shrinks target variance, so the predictors stay locally useful exactly where XOR lives.

4 Protocols divide the labour — which tells experimentalists what to run



Seven feature subsets × eight parameters. Left: normalised RMSE (lower better). Right: R^2 (higher better).

The full 94-feature model wins on average, but **no single protocol dominates every target**.

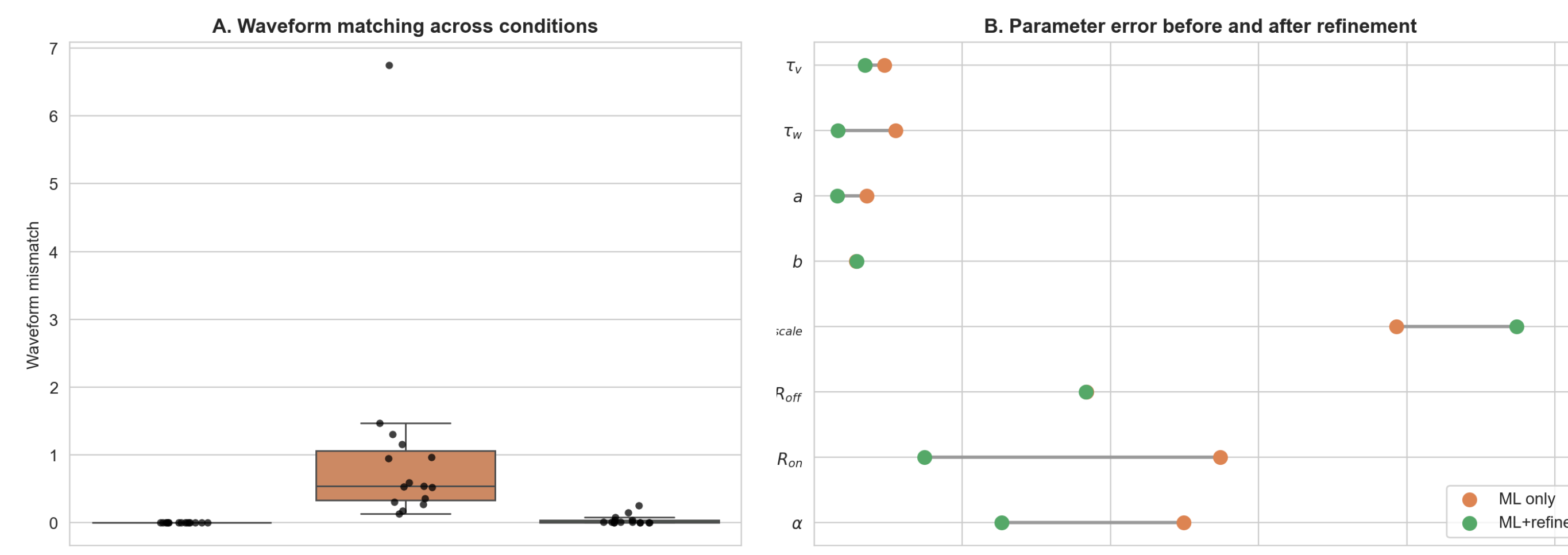
01 Step carries τ_v almost alone — R^2 0.83 step-only vs 0.87 with everything.

02 Paired-pulse is the single best addition: a jumps **0.46** → **0.69**. Short-timescale recovery is diagnostic of the excitation threshold.

03 Step + triangle is best for τ_w (0.75) and α (0.37) — slow recovery and memristance need a transient and a hysteretic probe.

Deployment order: step response first, paired-pulse second, triangle sweep third. v_{scale} and the resistance pair stay hard under every ablation — the present library underconstrains them.

5 Waveform-guided refinement rescues the ML guess



Mismatch by condition, $n = 15$.

Per-parameter error, **ML only** → **ML+refine**.

15 optimized specimens (20–50 nodes), the true graph copied into the surrogate. Dual annealing, bounds $\pm 20\%$ of range around the RF prediction.

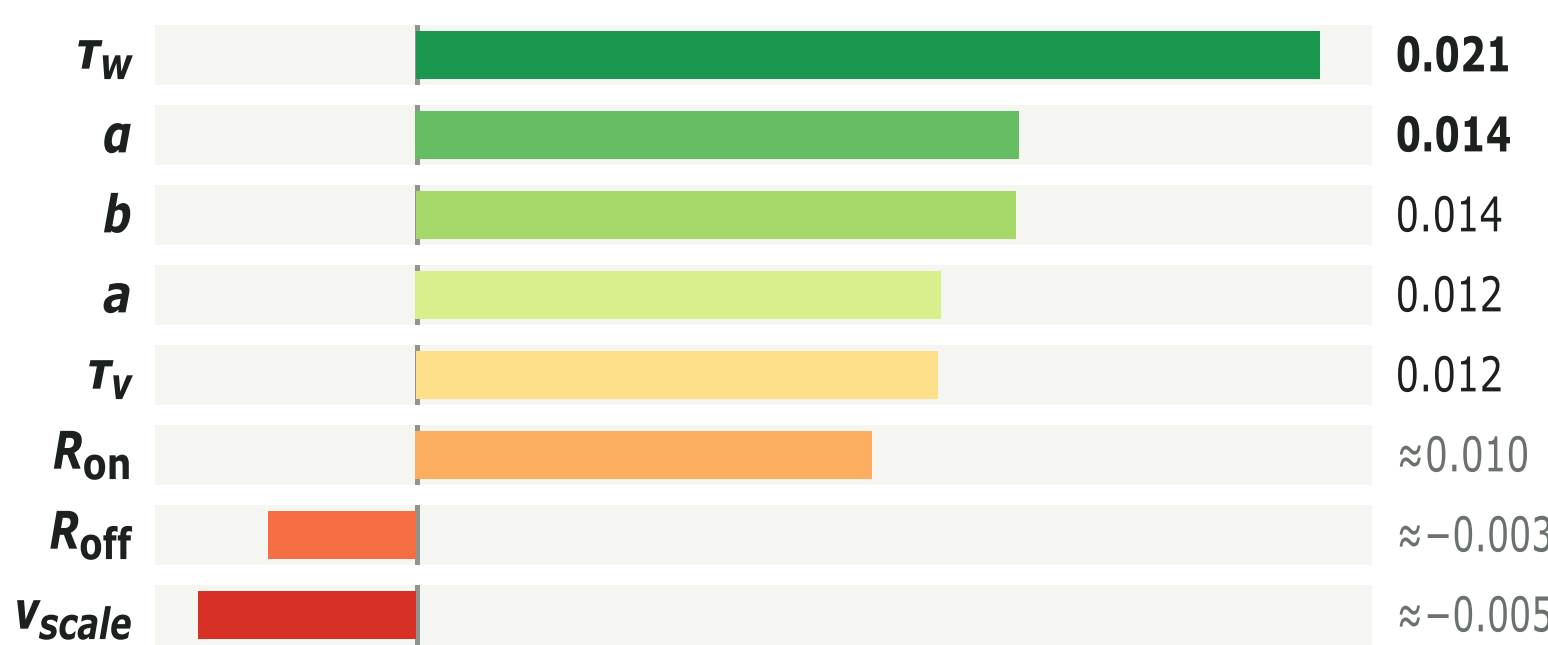
Condition	Mismatch	Err. all	Err. core
Oracle	0.000 ± 0.000	0.0 ± 0.0	0.0 ± 0.0
ML only	1.070 ± 1.625	31.6 ± 18.2	16.6 ± 5.9
ML+refine	0.042 ± 0.071	23.8 ± 20.3	8.8 ± 4.4

Where refinement bites ($p < 2.2 \times 10^{-4}$ each): τ_w 10.98→3.20%, a 7.04→3.08%, R_{on} 54.83→14.87%, α 49.90→25.33%.

Where it doesn't: τ_v improves but not significantly at this n ; b and R_{off} are unchanged; v_{scale} **worsens** (78.58→94.81%) — the objective simply cannot see it.

6 A sensitivity hierarchy that matches what we can measure

Mean XOR accuracy drop at $\pm 30\%$ perturbation, 72 viable specimens — redrawn from the paper's sensitivity figure for legibility.



Each parameter perturbed alone at $\pm 5/10/20/30\%$. Drops grow monotonically with magnitude, but per-specimen spread is wide enough that the confidence bands overlap across all eight parameters — *theranking* is the result, not the absolute drop.

The reassuring alignment. The parameters the workflow recovers best (τ_v, τ_w, a) are among the most consequential for XOR; the ones it recovers worst (v_{scale}, R_{off}) are where error is tolerable. Highest fidelity where it matters.

One exception names the next experiment. α is only moderately identifiable (R^2 0.396) yet ranks second in consequence — the memristive adaptation rate is the **highest-priority target for new protocol design**. And τ_w must be both correctly chosen *and* precisely estimated: the pivot of the whole pipeline.

SCOPE All validation is internal to the simulation

No physical specimens were tested. Margins, tuned parameters and optimized electrode geometries are **not claimed to transfer to living fungi**. Read this as a preparatory modelling and identifiability study.

Sampling. Characterization sampled the full parameter box, not the viable subset — viable-conditioned retraining is next.

Transfer. No test of whether a gate designed *in silico* works on an unknown specimen — the stronger end-to-end claim.

Topology. The specimen graph was copied into the surrogate, so this isolates parameter calibration, not structure inference.

Sensitivity design. One-at-a-time perturbation: no interaction effects, no input noise, no electrode jitter.

What we build next

Conductance-sensitive probes. Impedance spectroscopy and frequency-dependent stimulation to resolve a, v_{scale}, R_{off} .

Posteriori, not point estimates — plus Bayesian optimization of the protocol itself: pick the next probe from what stays uncertain.

Hidden-topology inference. Scored structurally (graph-edit, spectral) *and* functionally (gate transfer to held-out specimens).

Beyond XOR. Any task scorable from network output, including reservoir capacity — then physical validation.